

JACK’S RESEARCH LAB
Research Note 004

Did Remote Work Increase Productivity?

Evidence from U.S. Industries, 2019–2023

Jack Shaffer, M.A. Economics
June 2026

Research Question: Did industries that adopted remote work experience stronger productivity growth, or were remote-work industries already different?

Abstract

Remote work became one of the defining labor market changes of the COVID-19 era. Some observers argue that remote work improved productivity, while others argue that it weakened coordination, mentoring, and workplace efficiency. This note examines a more subtle question: whether industries that adopted remote work became more productive, or whether remote work was concentrated in industries that were already different.

Using 2019 and 2023 American Community Survey microdata from IPUMS USA and Bureau of Labor Statistics industry productivity data, I compare remote-work adoption, worker education, and labor productivity growth across 38 matched three-digit NAICS industries. The results show that remote-work adoption was strongly concentrated in highly educated industries. However, remote-work adoption was not meaningfully associated with faster industry-level labor productivity growth from 2019 to 2023.

Study Design

Research Question	Did remote-work industries experience faster productivity growth?
Period	2019–2023
Unit of Analysis	Three-digit NAICS industry
Remote Work Measure	Change in work-from-home share
Productivity Measure	BLS labor productivity index growth
Control Variable	2019 college share by industry
Final Sample	38 matched industries

Remote work is measured using the ACS means-of-transportation-to-work variable, where workers reporting “worked at home” are coded as remote workers. Education is measured as the share of employed workers with a bachelor’s degree or higher. Labor productivity growth is calculated from BLS industry labor productivity indexes:

$$ProductivityGrowth_i = \frac{Productivity_{i,2023}}{Productivity_{i,2019}} - 1$$

Remote-work adoption is measured as:

$$RemoteChange_i = WFHShare_{i,2023} - WFHShare_{i,2019}$$

Empirical Model

The baseline model estimates the relationship between remote-work adoption and labor productivity growth:

$$ProductivityGrowth_i = \beta_0 + \beta_1 RemoteChange_i + \epsilon_i$$

A second specification adds worker education:

$$ProductivityGrowth_i = \beta_0 + \beta_1 RemoteChange_i + \beta_2 CollegeShare_{i,2019} + \epsilon_i$$

A third specification adds the pre-pandemic work-from-home share:

$$ProductivityGrowth_i = \beta_0 + \beta_1 RemoteChange_i + \beta_2 CollegeShare_{i,2019} + \beta_3 WFHShare_{i,2019} + \epsilon_i$$

These models should be interpreted as exploratory cross-industry regressions, not causal estimates. The purpose is to test whether the simple relationship between remote work and productivity survives basic controls for industry composition.

Main Results

The first result is that remote-work adoption was not random across industries. Industries with more educated workforces in 2019 adopted remote work much more aggressively from 2019 to 2023.

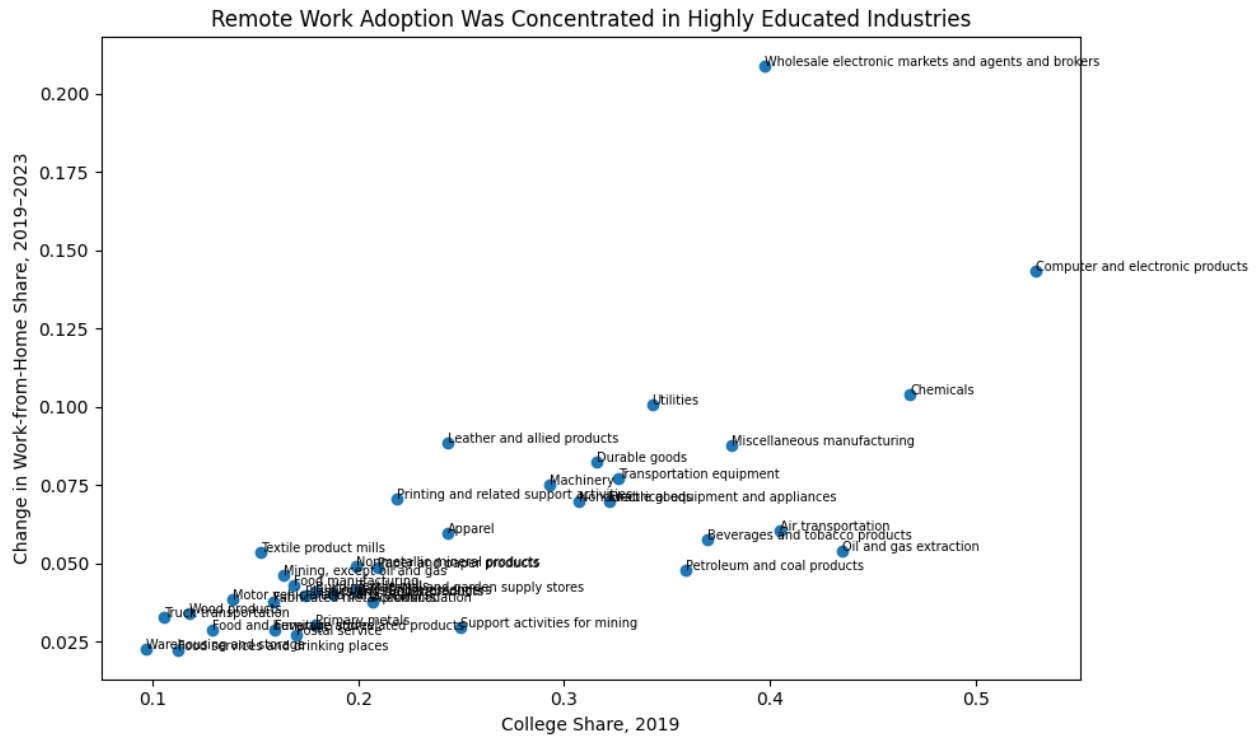


Figure 1: Remote Work Adoption and College Share by Industry

The correlation between 2019 college share and remote-work adoption was:

$$r = 0.721$$

with a p-value below 0.0001. This means that industries with more college-educated workers were substantially more likely to shift toward remote work.

However, remote-work adoption was not meaningfully associated with labor productivity growth. The correlation between remote-work adoption and productivity growth was:

$$r = 0.078$$

with a p-value of 0.643.

Model	Remote Coefficient	Remote p-value	R^2	N
Remote only	0.243	0.519	0.006	38
Add college share	-0.036	0.966	0.013	38
Add pre-pandemic WFH	-0.368	0.759	0.017	38

The regression results provide little evidence that industries with larger increases in remote work experienced faster productivity growth. The remote-work coefficient is statistically insignificant in all three specifications.

Interpretation

The key finding is subtle but important. Remote work was not randomly distributed across the economy. It expanded most in industries with highly educated, white-collar, knowledge-intensive workforces. This creates a major confounding issue for simple claims about remote work and productivity.

A naive comparison might observe that remote-heavy industries are productive and conclude that remote work caused higher productivity. But that conclusion skips an important step. These industries were already different before the pandemic. They had more educated workforces, more digital work processes, and jobs that were easier to perform outside a traditional office.

In this sample, remote-work adoption is strongly related to education, but not to productivity growth. That suggests caution toward both overly positive and overly negative claims about remote work. The data do not show that remote work clearly boosted industry productivity. But they also do not show that remote work damaged productivity at the industry level.

Key Findings

- Remote work adoption was highly uneven across industries.
- Industries with more educated workforces adopted remote work more aggressively.
- The correlation between college share and remote-work adoption was strong: $r = 0.721$.
- Remote-work adoption was not meaningfully correlated with labor productivity growth: $r = 0.078$.
- In regression models, the remote-work coefficient was statistically insignificant with and without controls.

Limitations

This analysis should be interpreted as exploratory. Several limitations are important:

- The analysis is conducted at the industry level, not the firm or worker level.
- ACS work-from-home status measures usual work location and may not fully capture hybrid work.
- Industry productivity measures may reflect many factors unrelated to remote work, including capital investment, demand shocks, automation, supply chains, and energy prices.
- The matched productivity sample contains 38 industries, which is useful for exploration but limited for strong causal claims.
- The analysis does not identify whether remote work improved productivity for some individual workers or firms.

A stronger causal study would use firm-level or worker-level data, better measures of hybrid work, and a design that isolates remote-work adoption from industry composition.

Data Sources

Remote-work and education measures were constructed from 2019 and 2023 American Community Survey microdata accessed through IPUMS USA. Remote work was identified using the ACS transportation-to-work variable, where “worked at home” indicates remote work. Education was measured using detailed educational attainment, with bachelor’s degree or higher coded as college educated.

Labor productivity data were obtained from the Bureau of Labor Statistics Office of Productivity and Technology detailed industry productivity dataset. Productivity growth was calculated using the labor productivity index for all workers.

Conclusion

Remote work did not spread randomly across the economy. It was concentrated in industries that were already more educated and more suited to digital work. That matters because it complicates simple claims about remote work and productivity.

In this industry-level sample, remote-work adoption was strongly associated with workforce education, but not with faster productivity growth from 2019 to 2023. The best interpretation is not that remote work obviously increased productivity, nor that it obviously reduced it. Instead, the evidence suggests that remote work was heavily shaped by pre-existing industry differences.

The productivity story is therefore more complicated than the headline debate suggests. Remote work may change how people work, but industries that went remote were not randomly selected. They were already different.

JACK'S RESEARCH LAB
Economics – Data Science – Forecasting – Curiosity
Exploring interesting questions with data.

Author: Jack Shaffer, M.A. Economics
University of Wisconsin–Milwaukee
Jack's Research Lab